

Forearm

Department of anatomy

Musculoskeletal System

Third Part

The Upper Limb (6)

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Lecture points

- ✓ Introduction
- ✓ Compartments.
- Muscles of the Forearm
- ✓ Contents of the Anterior Fascial Compartment of the Forearm
 - ✓ Muscles.
- ✓ Contents of the Posterior Fascial Compartment of the Forearm.
 - ✓ Muscles.
- Blood supply in the forearm:
 - Ulnar artery in the forearm.
 - Anterior interosseous artery.
 - Posterior interosseous artery.
 - Radial artery in the forearm.
 - Surface Marking of ulnar and radial artery.
- ✓ Summary of arteries of the Anterior and Posterior Compartment of the Forearm
- ✓ Venous drainage in the arm and Forearm
- ✓ Lymphatic drainage in the forearm
- Innervation in the Forearm.
- ✓ Median nerve in the anterior fascial compartment of the forearm.
- ✓ Anterior Interosseous Nerve .
- ✓ Ulnar nerve in the anterior fascial compartment of the forearm.
- ✓ Radial Nerve and the posterior interosseous nerve of the Forearm
- ✓ Innervation :
 - ✓ Sensory (dermatome).
 - ✓ Motor (myotome).

Innervation of forearm

Median nerve

Ulnar nerve

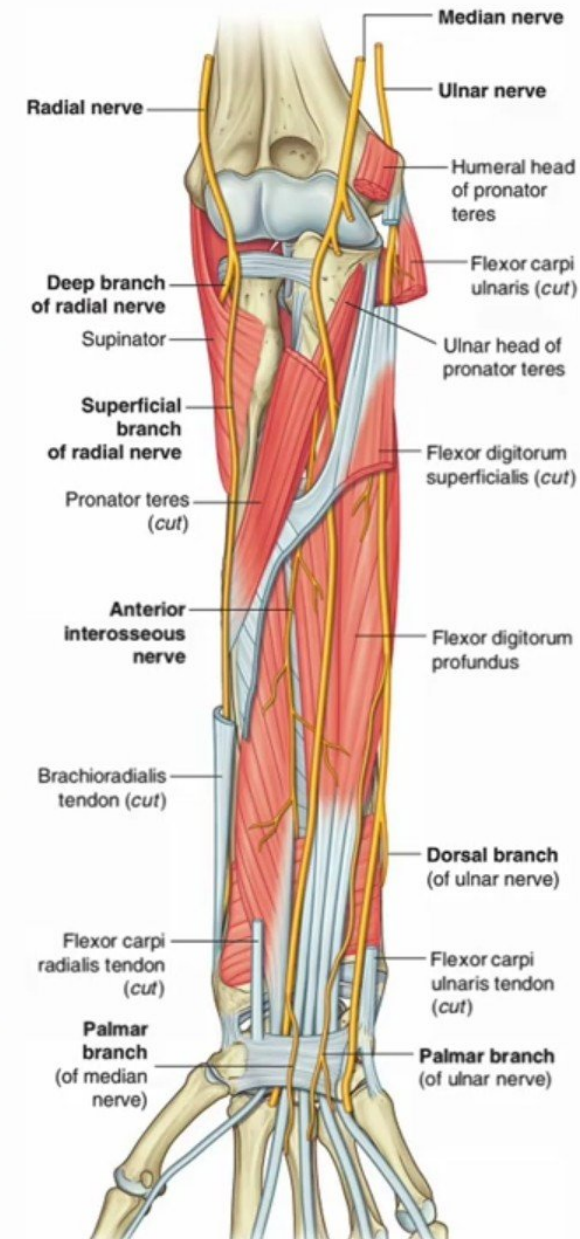
Radial nerve



Median nerve in the anterior fascial compartment of the forearm

Median Nerve:

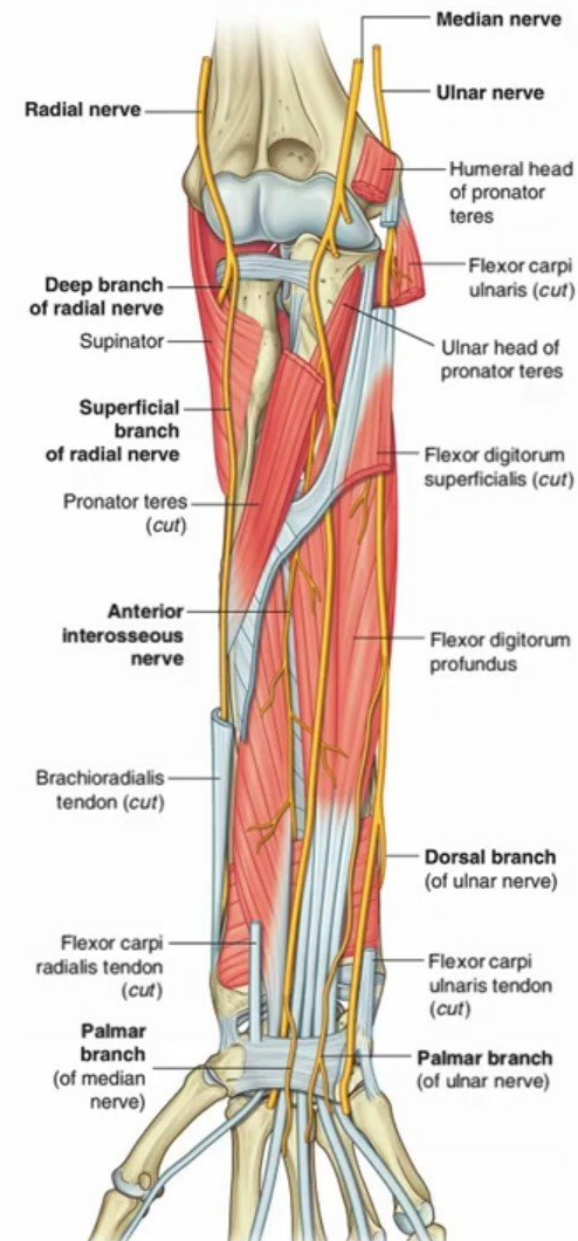
- It leaves the cubital fossa by passing between **the two heads of the pronator teres**.
- It continues downward **behind the flexor digitorum superficialis** and rests **posteriorly on the flexor digitorum profundus**.
- **At the wrist**, the median nerve emerges from the **lateral border of the flexor digitorum superficialis muscle** and lies **behind the tendon of the palmaris longus**.
- It enters the palm by passing **inferior to** the flexor retinaculum.



Median nerve in the anterior fascial compartment of the forearm

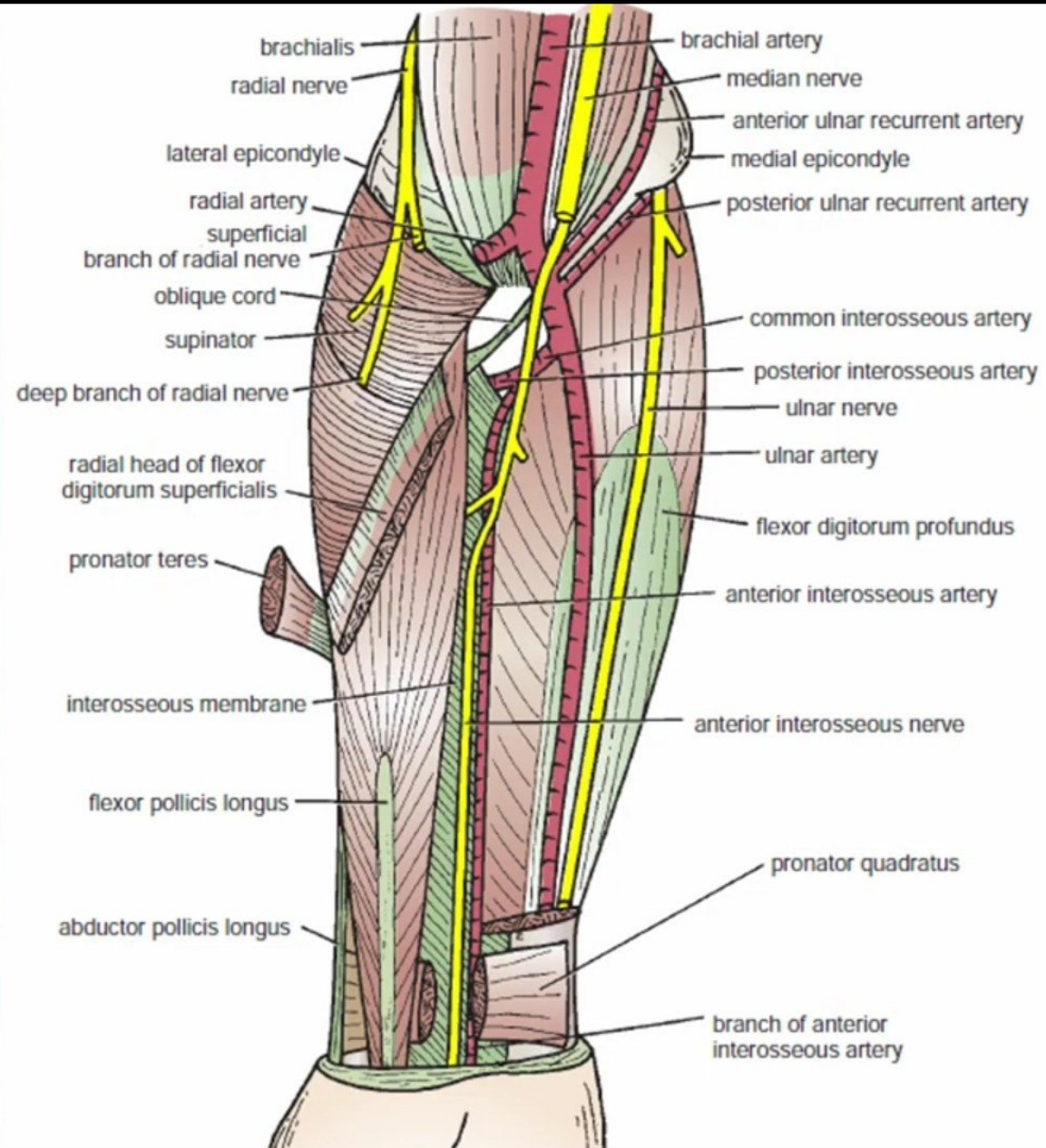
Median Nerve branches:

- **Muscular branches** in the cubital fossa to the pronator teres, the flexor carpi radialis, the palmaris longus, and the flexor digitorum superficialis
- **Articular branches** to the elbow joint
- **Anterior interosseous nerve**
- **Palmar cutaneous branch.**
- This arises in the lower part of the forearm and is distributed to the skin over the lateral part of the palm .

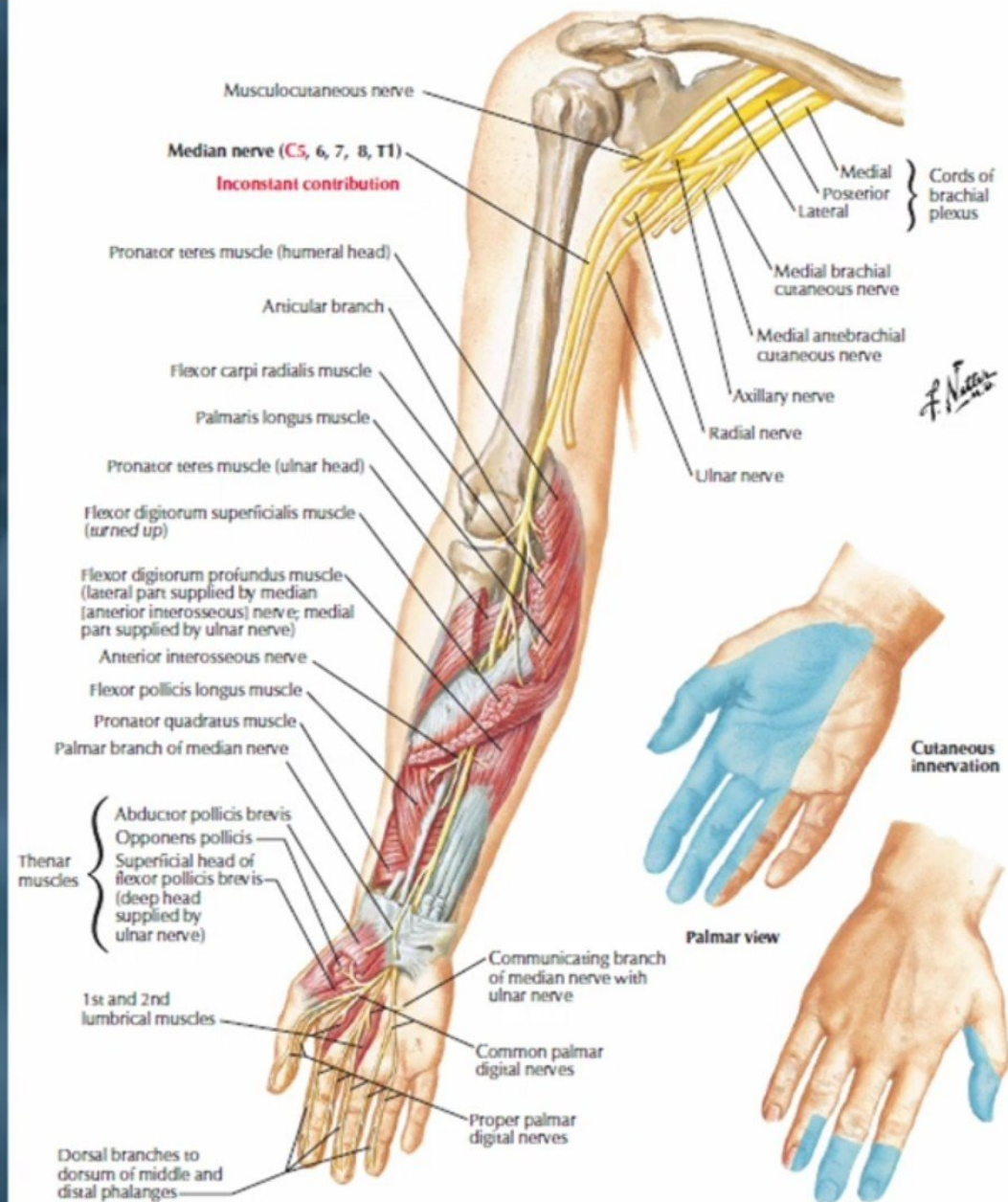
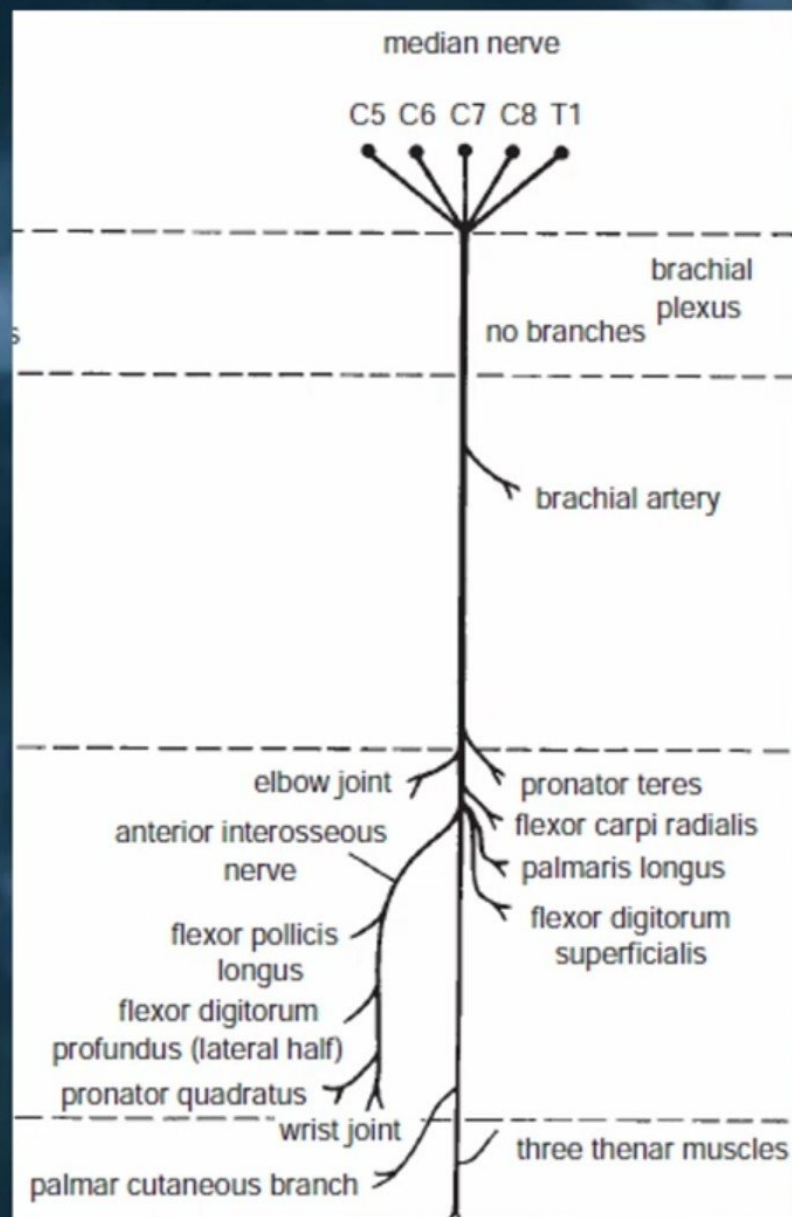


Anterior Interosseous Nerve

- It arises from **the median nerve** as it emerges from between the **two heads of the pronator teres**.
- It passes downward on **the anterior surface of the interosseous membrane**, between the **flexor pollicis longus** and the **flexor digitorum profundus**.
- It ends on the anterior surface of **the carpus**.
- **Branches:**
- ✓ **Muscular branches to:** the **flexor pollicis longus**, the **pronator quadratus**, and the **lateral half of the flexor digitorum profundus**.
- ✓ **Articular branches to:** the wrist, distal radioulnar joints and the joints of the hands.



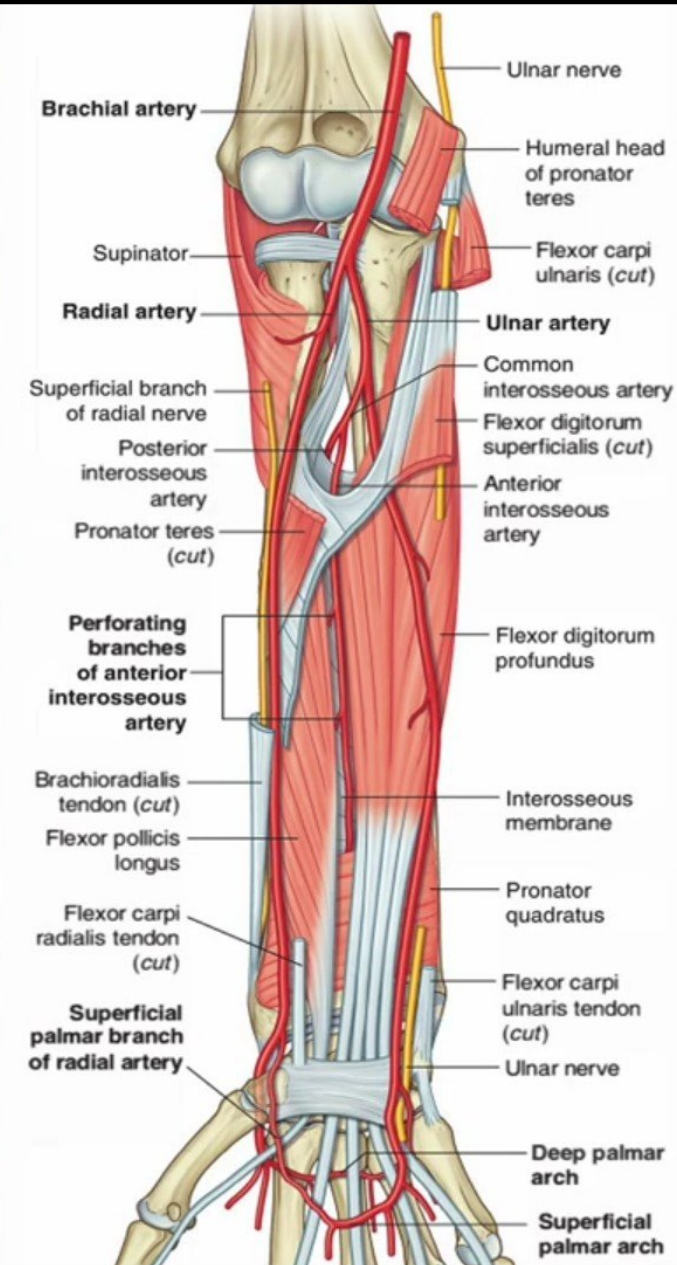
Summary of Median Nerve in forarm



Ulnar nerve in the anterior fascial compartment of the forearm

The ulnar nerve:

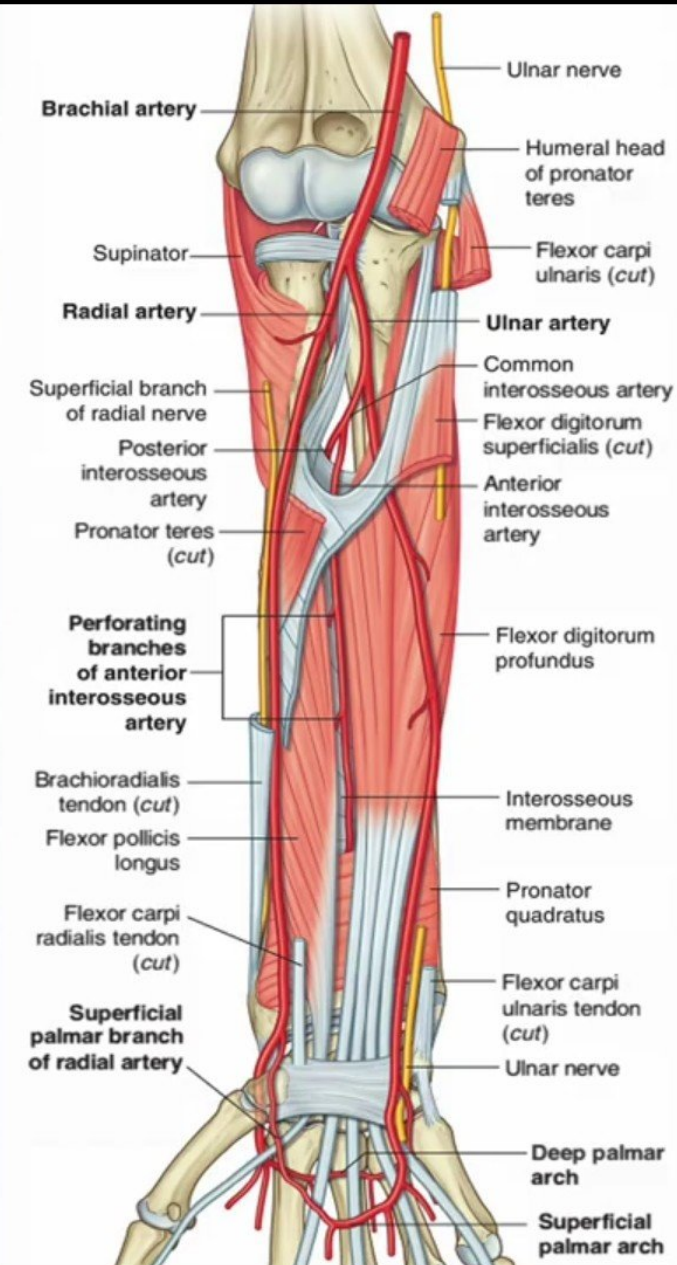
- It passes **behind the medial epicondyle** of the humerus, **crosses the medial ligament** of the elbow joint, and **enters the front of the forearm** by **passing between the two heads of the flexor carpi ulnaris**.
- It then **runs down** the forearm between the
- **flexor carpi ulnaris** and **the flexor digitorum profundus** muscles.
- In the **distal two thirds** of the forearm, **the ulnar artery** lies **on the lateral side of the ulnar nerve**.



Ulnar nerve in the anterior fascial compartment of the forearm

The ulnar nerve:

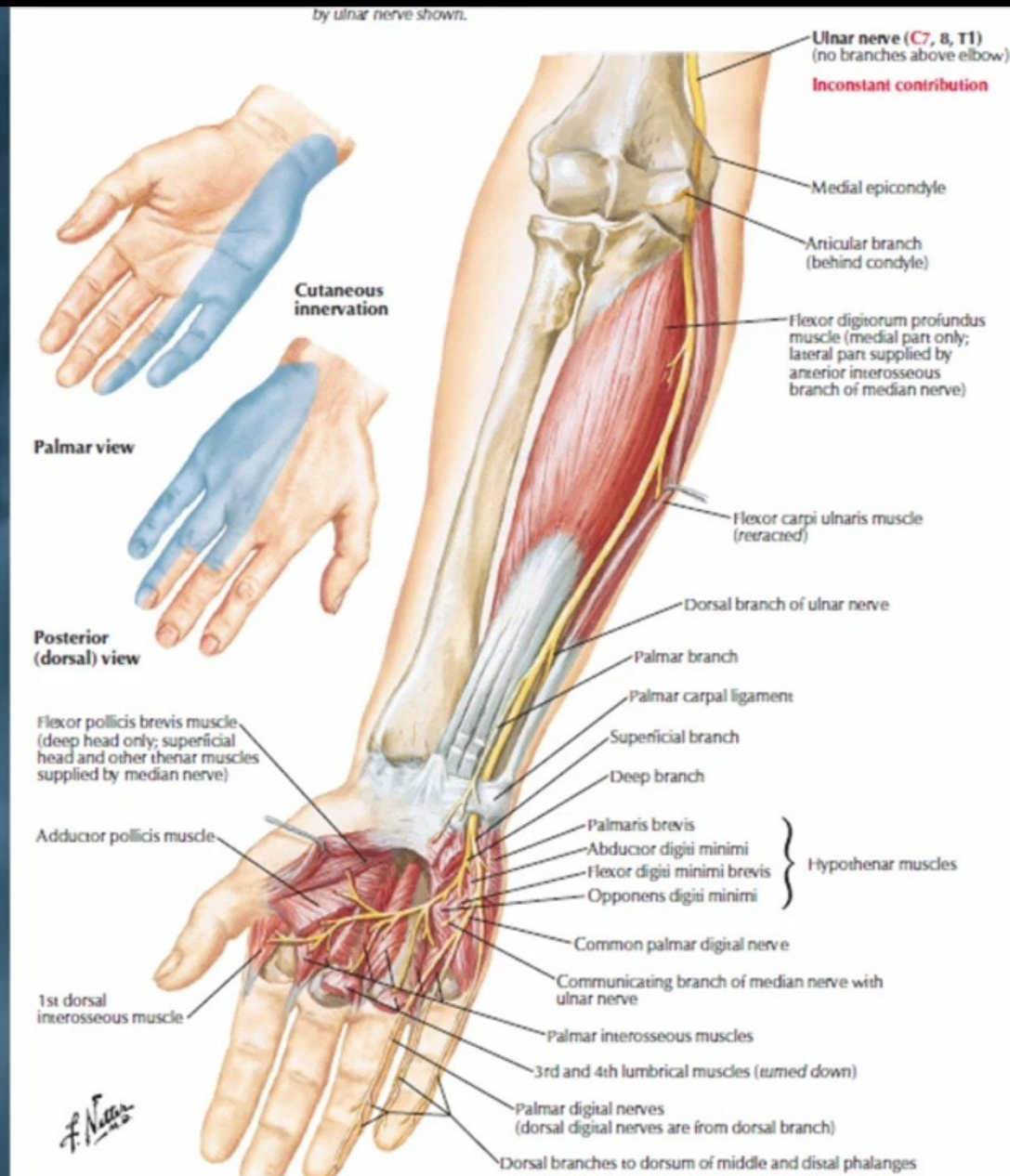
- **At the wrist**, the ulnar nerve **becomes superficial** and lies
- between **the tendons of the flexor carpi ulnaris** and **flexor**
- **digitorum superficialis muscles.**
- It enters the palm of the hand by passing **anterior to the flexor retinaculum** and **lateral to the pisiform bone.**
- it has the ulnar artery lateral to the ulnar nerve.



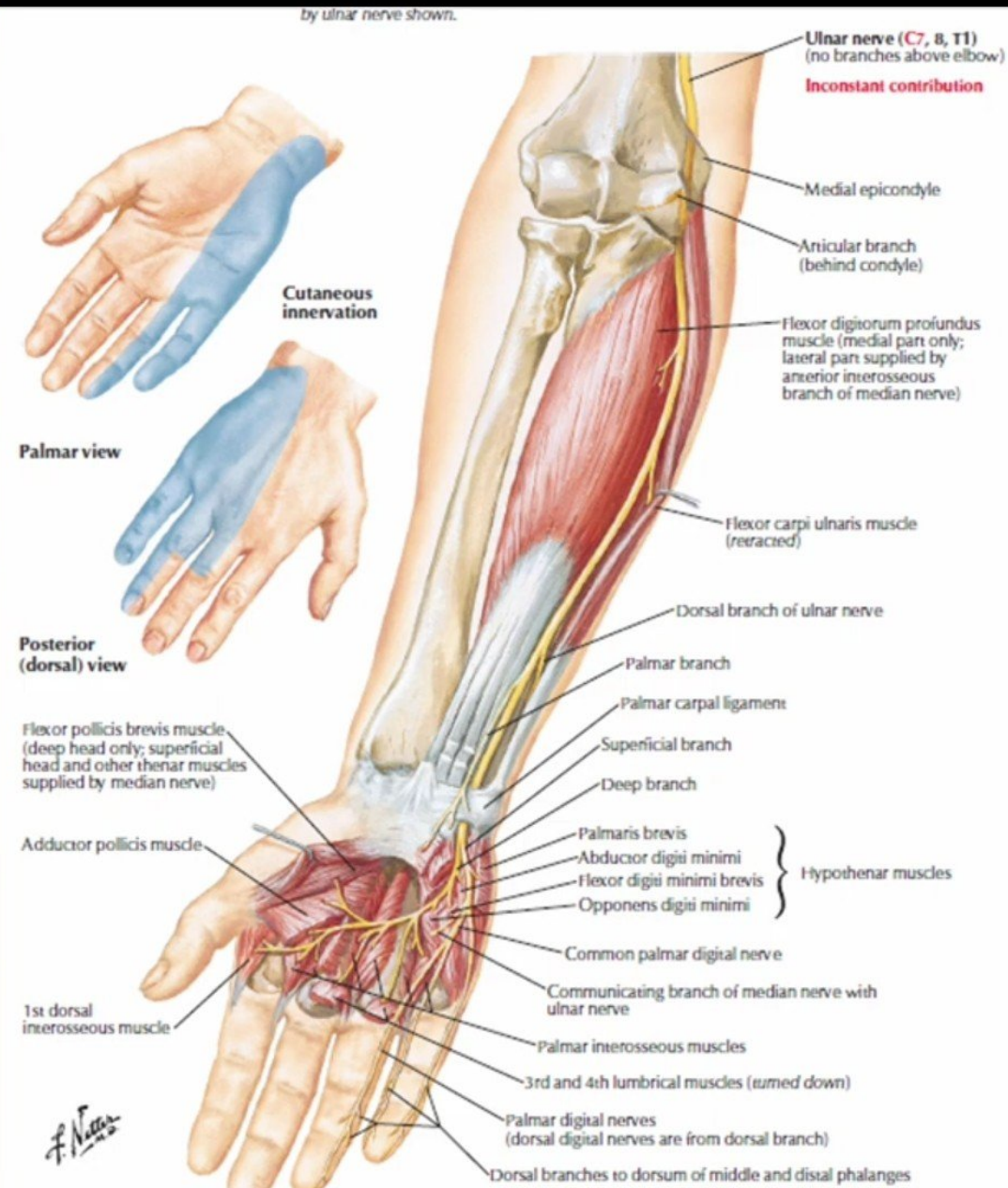
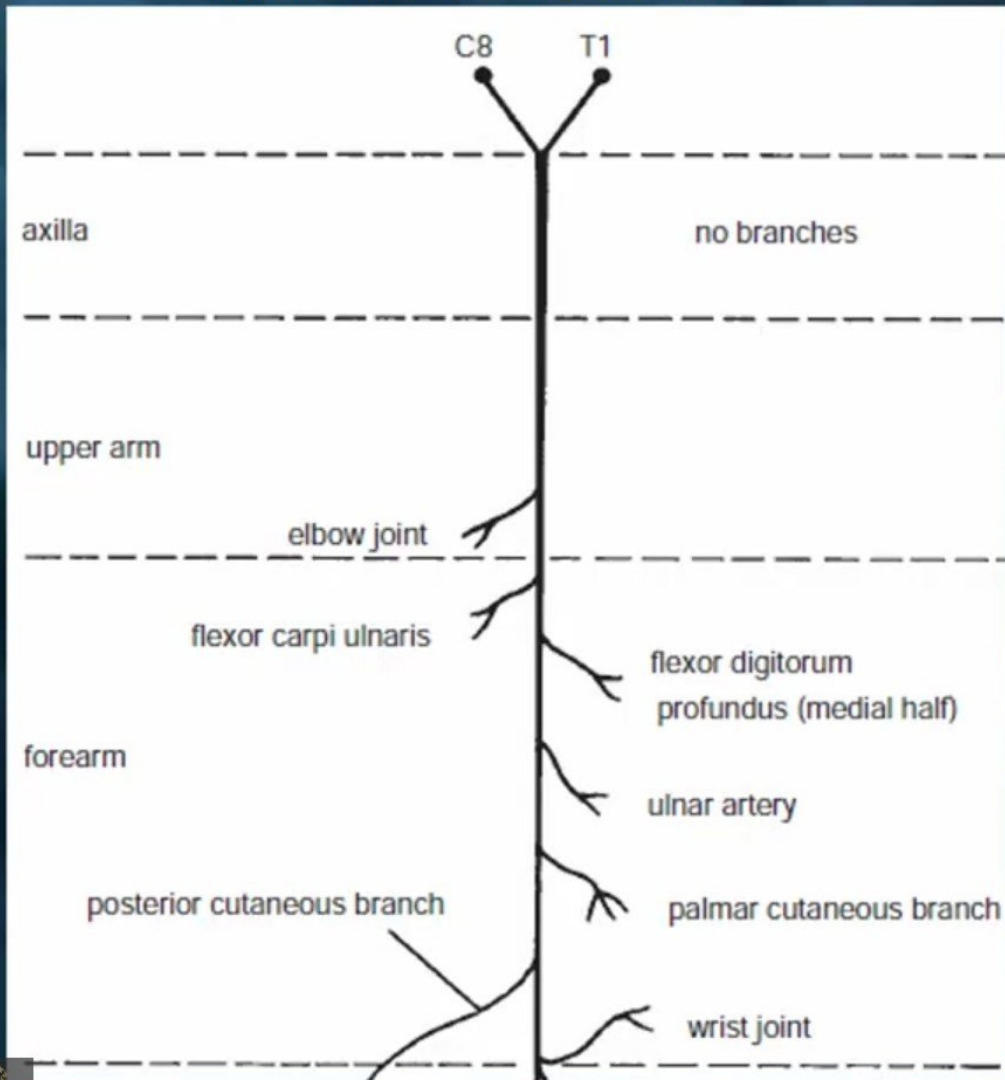
Ulnar nerve in the anterior fascial compartment of the forearm

Branches of the ulnar nerve:

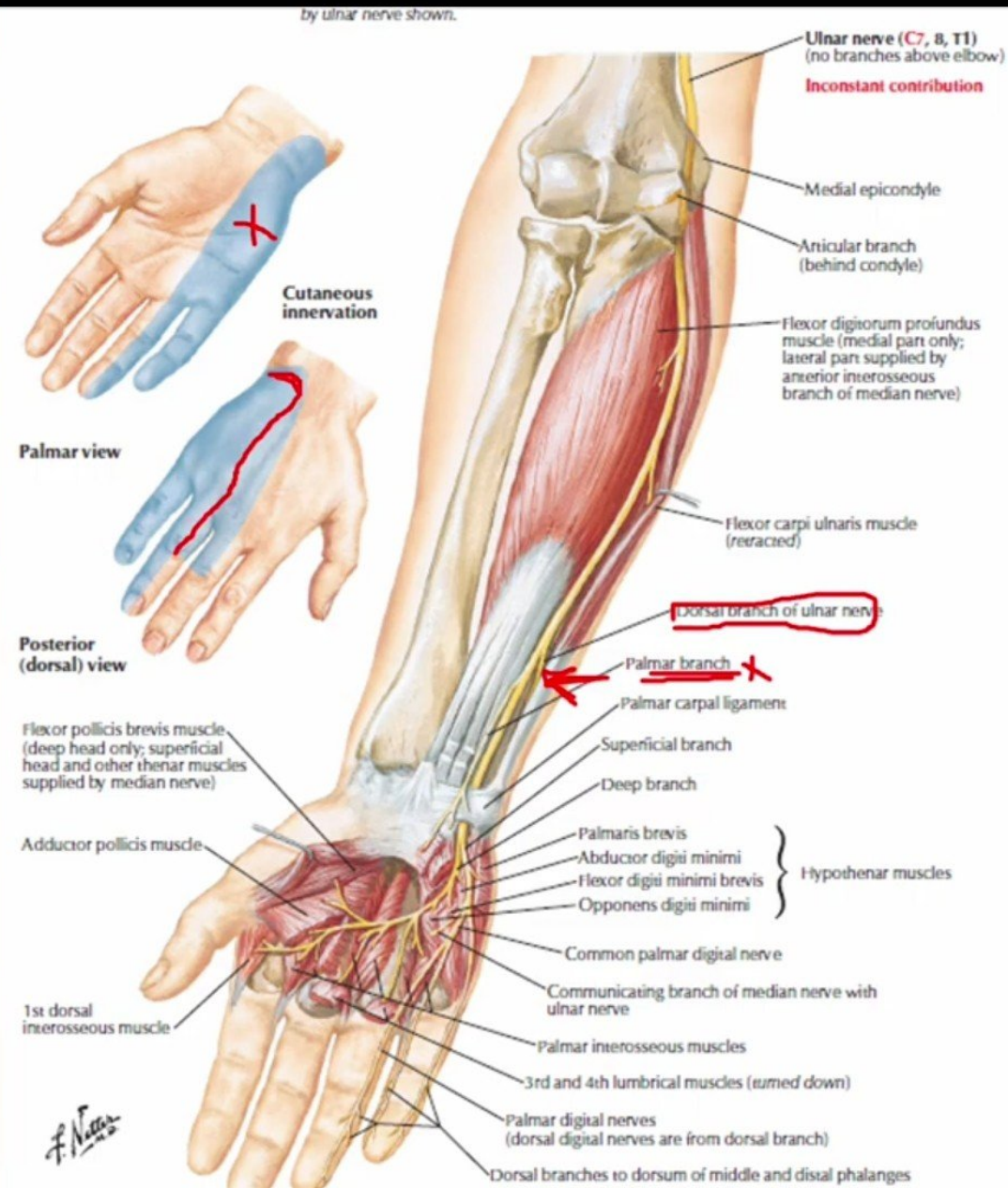
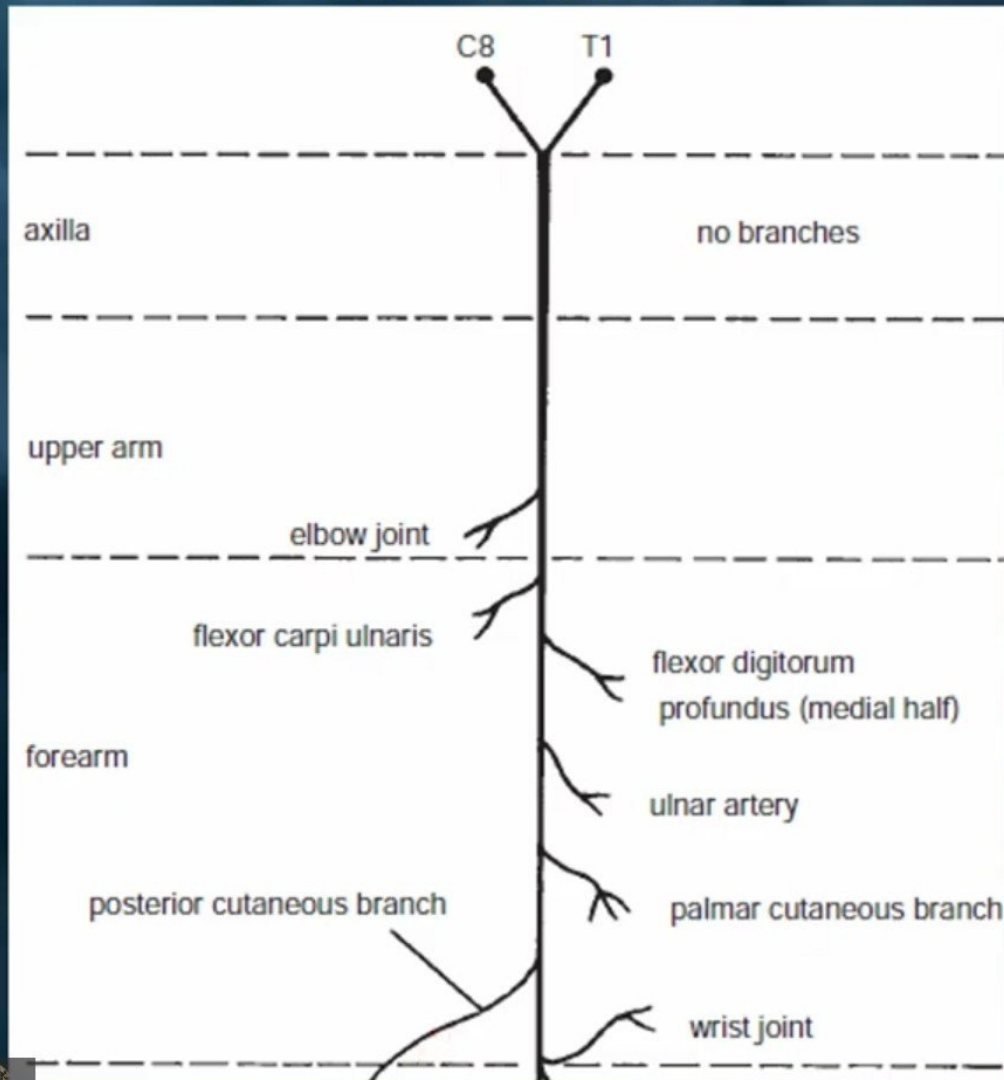
- **Muscular branches** to the flexor carpi ulnaris and the medial half of the flexor digitorum profundus.
- **Articular branches** to the elbow joint.
- **The palmar cutaneous branch** is a small branch that arises in the middle of the forearm and supplies the skin over the hypothenar eminence.
- **The dorsal posterior cutaneous branch**
 - It is a large branch that arises in the distal third of the forearm.
 - It passes medially between the tendon of the flexor carpi ulnaris and the ulna and
 - It is distributed on the posterior surface of the hand and fingers.



Summary of Ulnar Nerve in the forearm

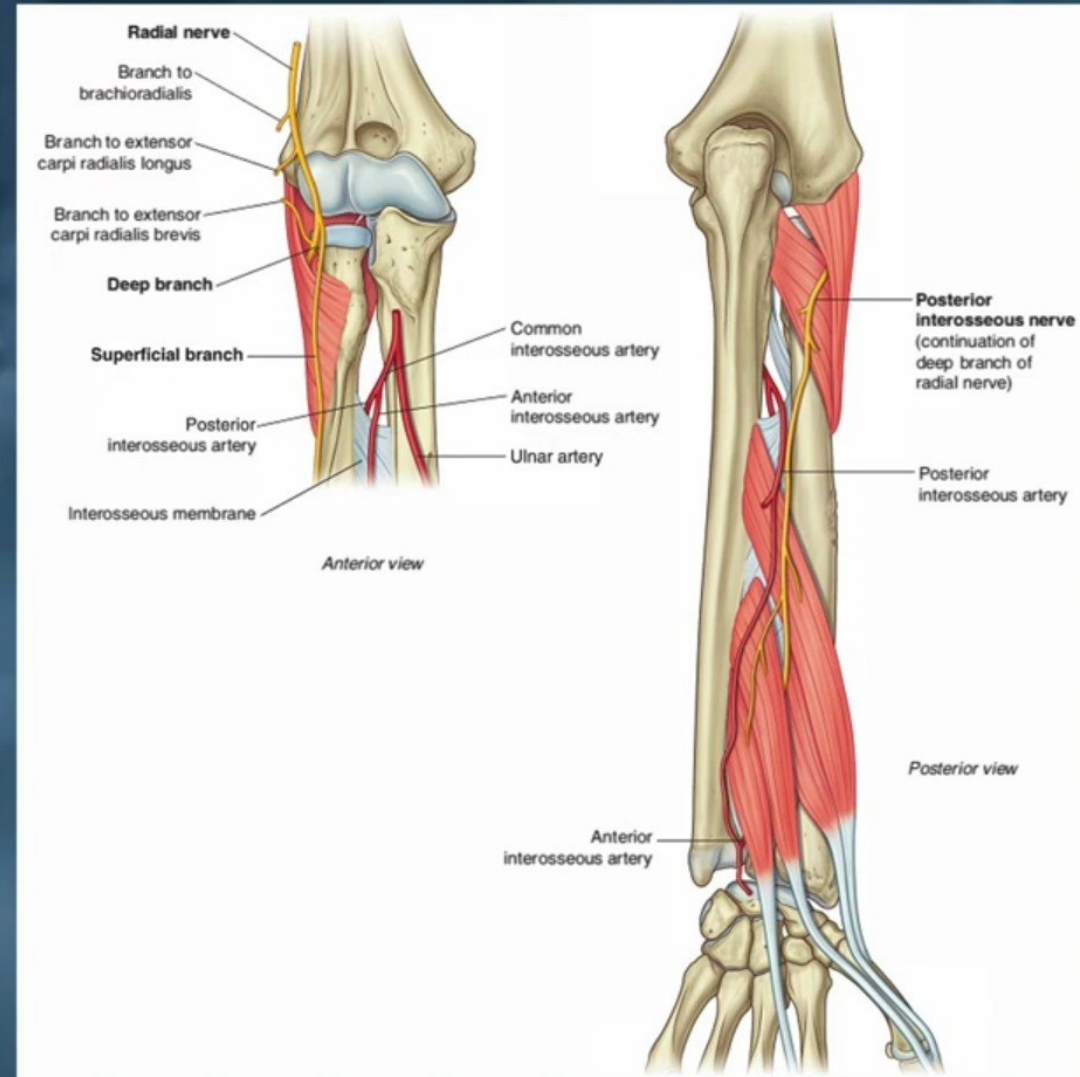


Summary of Ulnar Nerve in the forearm



Radial Nerve and the posterior interosseous nerve of the Forearm

- It is **deep Branch of the Radial Nerve (posterior interosseous nerve)**
- The radial nerve descend in front **of the lateral epicondyle of the humerus** in the cubital fossa.
- It pass between head of **the supinator** and winds around the lateral aspect of the neck of the radius to reach the **posterior compartment of the forearm**.
- This nerve named **posterior interosseous nerve** than descends in the interval between **the superficial and deep groups of muscles**.
- It eventually reaches the posterior surface of the wrist joint.



Radial Nerve and the posterior interosseous nerve of the Forearm

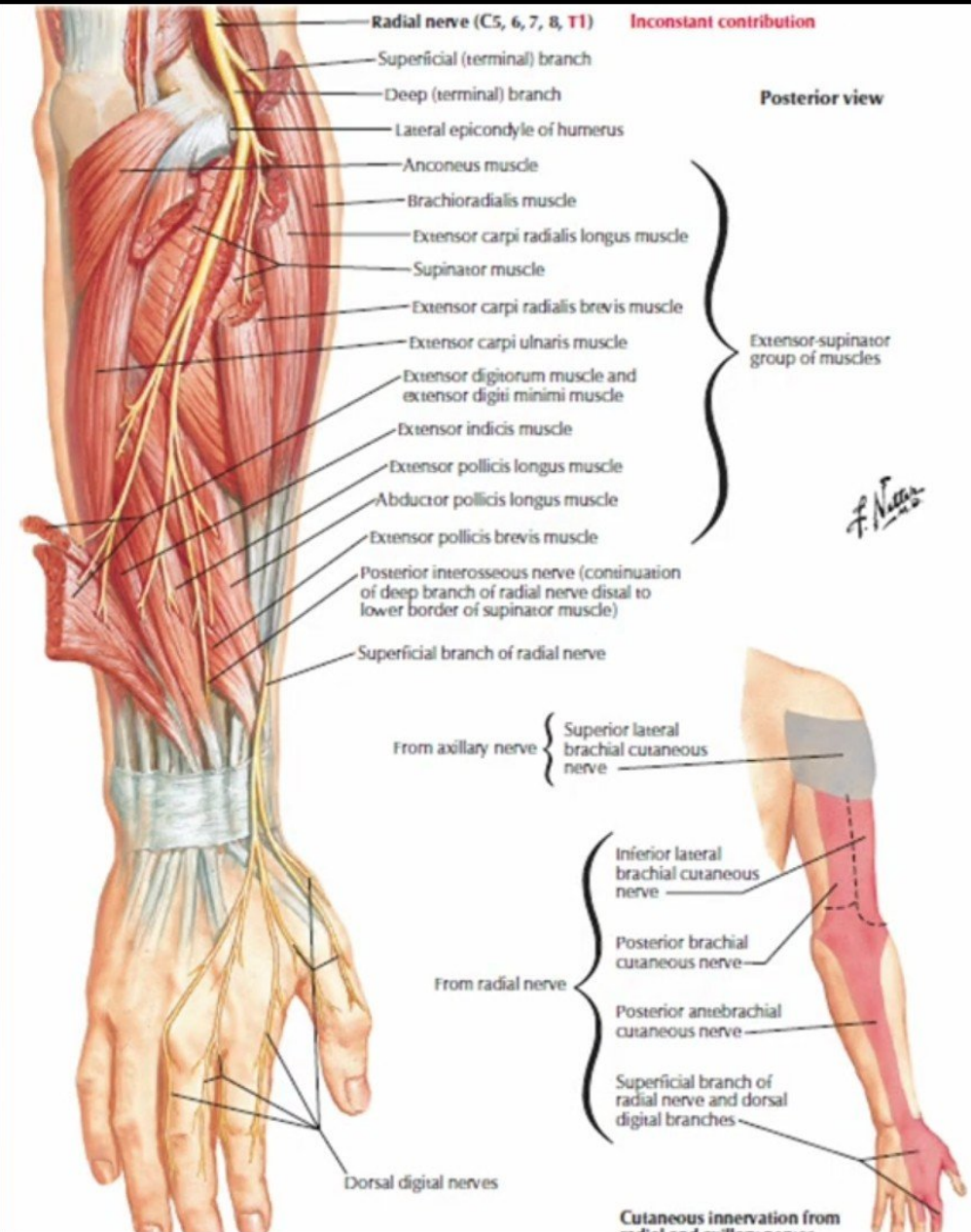
Branches

Muscular branches to:

- The extensor carpi radialis brevis and the supinator, the extensor digitorum, the extensor digiti minimi, the extensor carpi ulnaris, the abductor pollicis longus, the extensor pollicis brevis, the extensor pollicis longus, and the extensor indicis

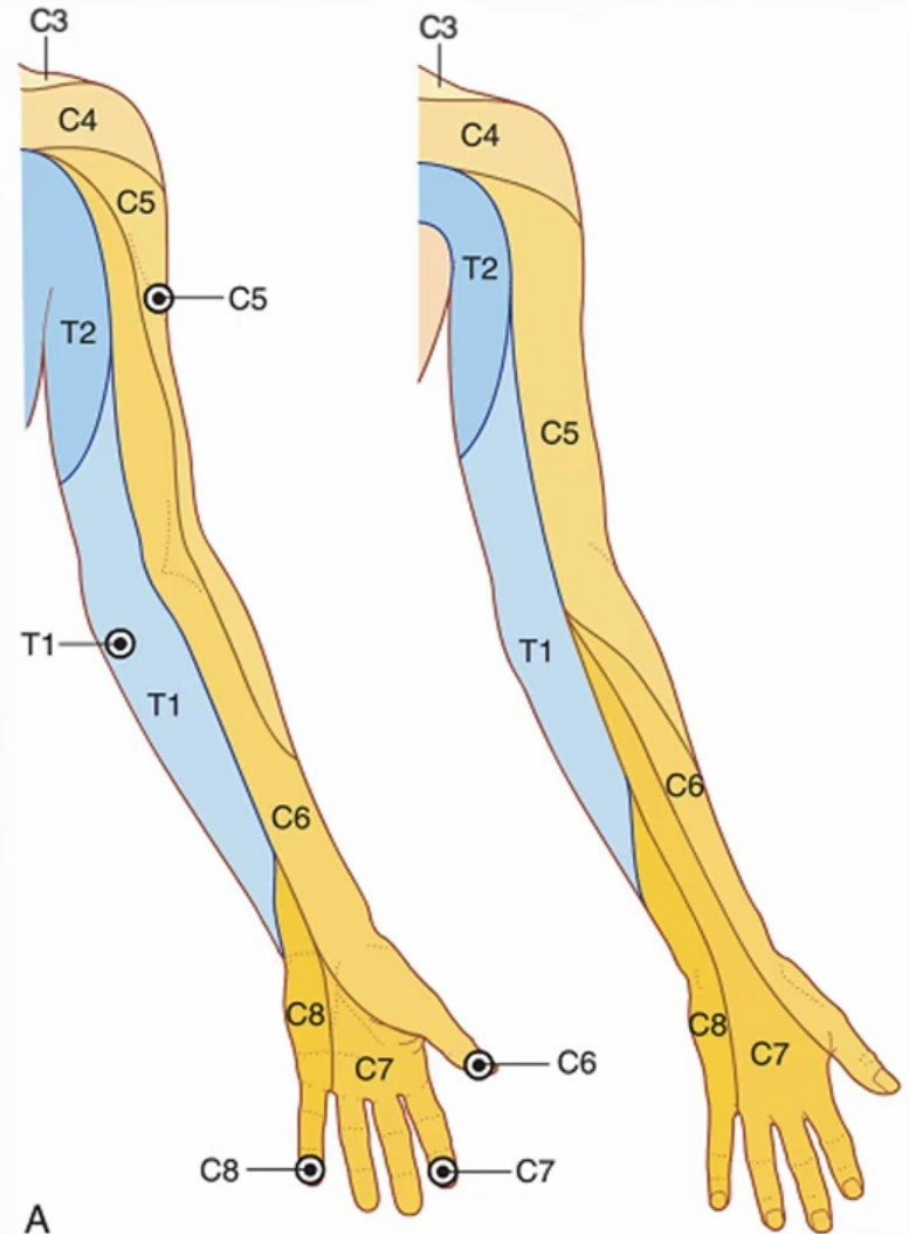
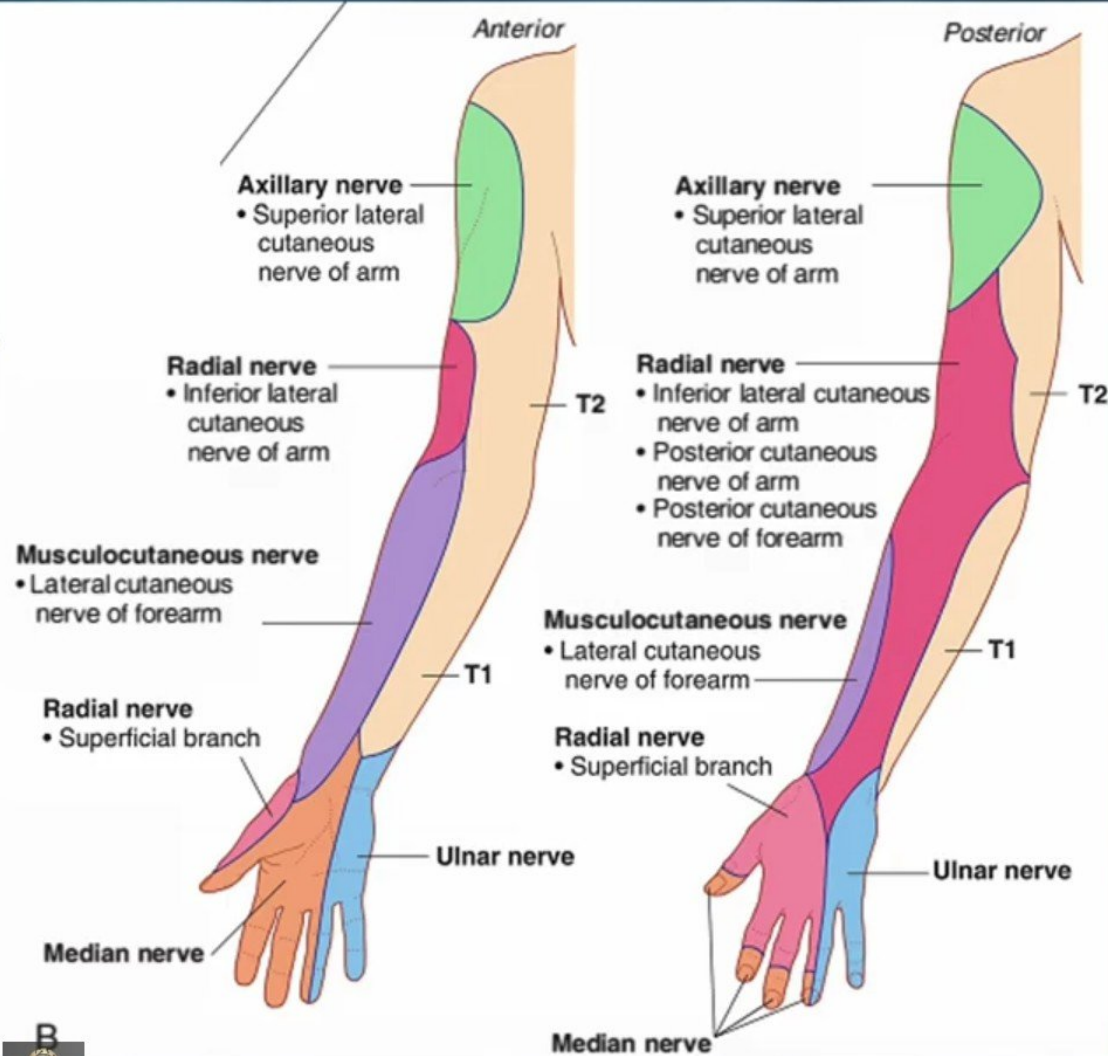
Articular branches to:

- The wrist and carpal joints.



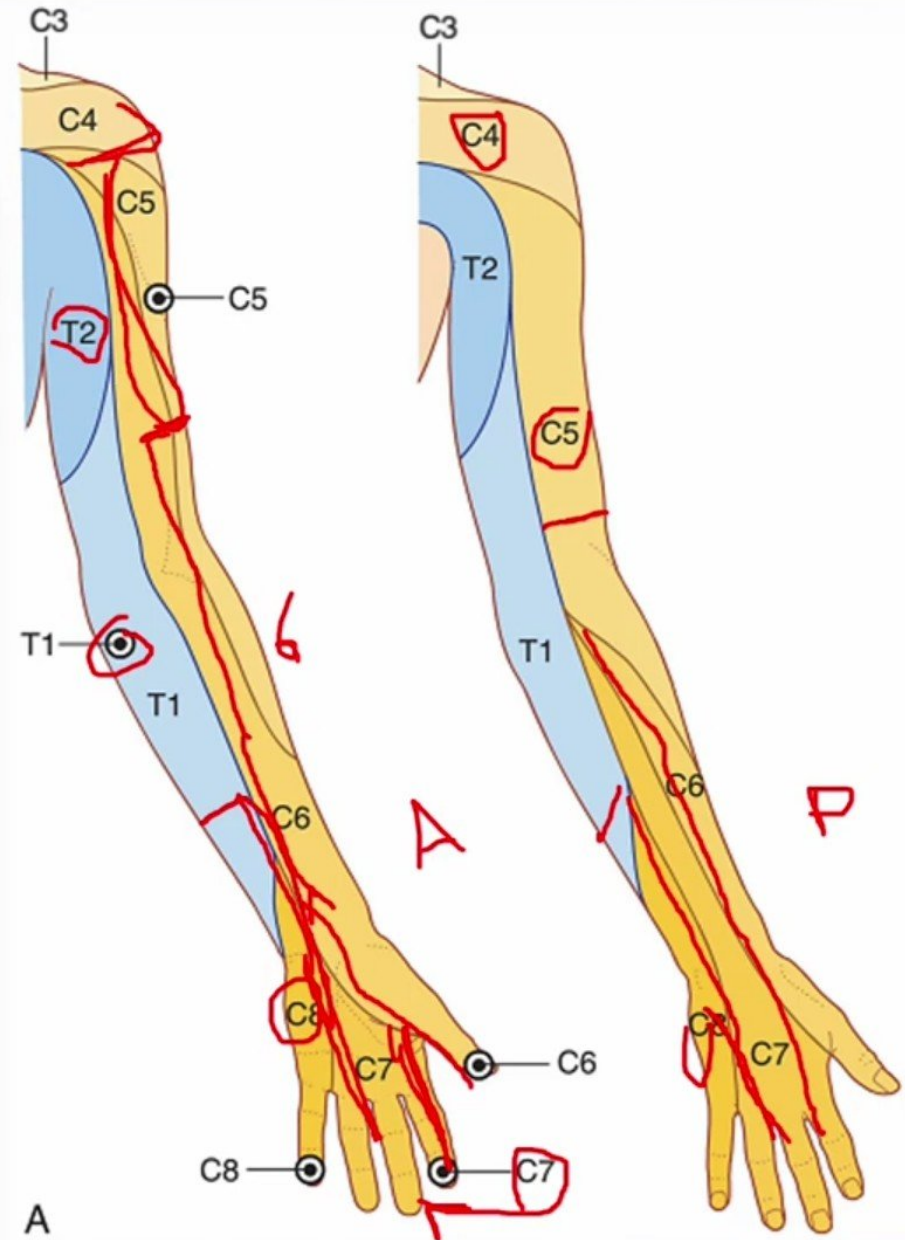
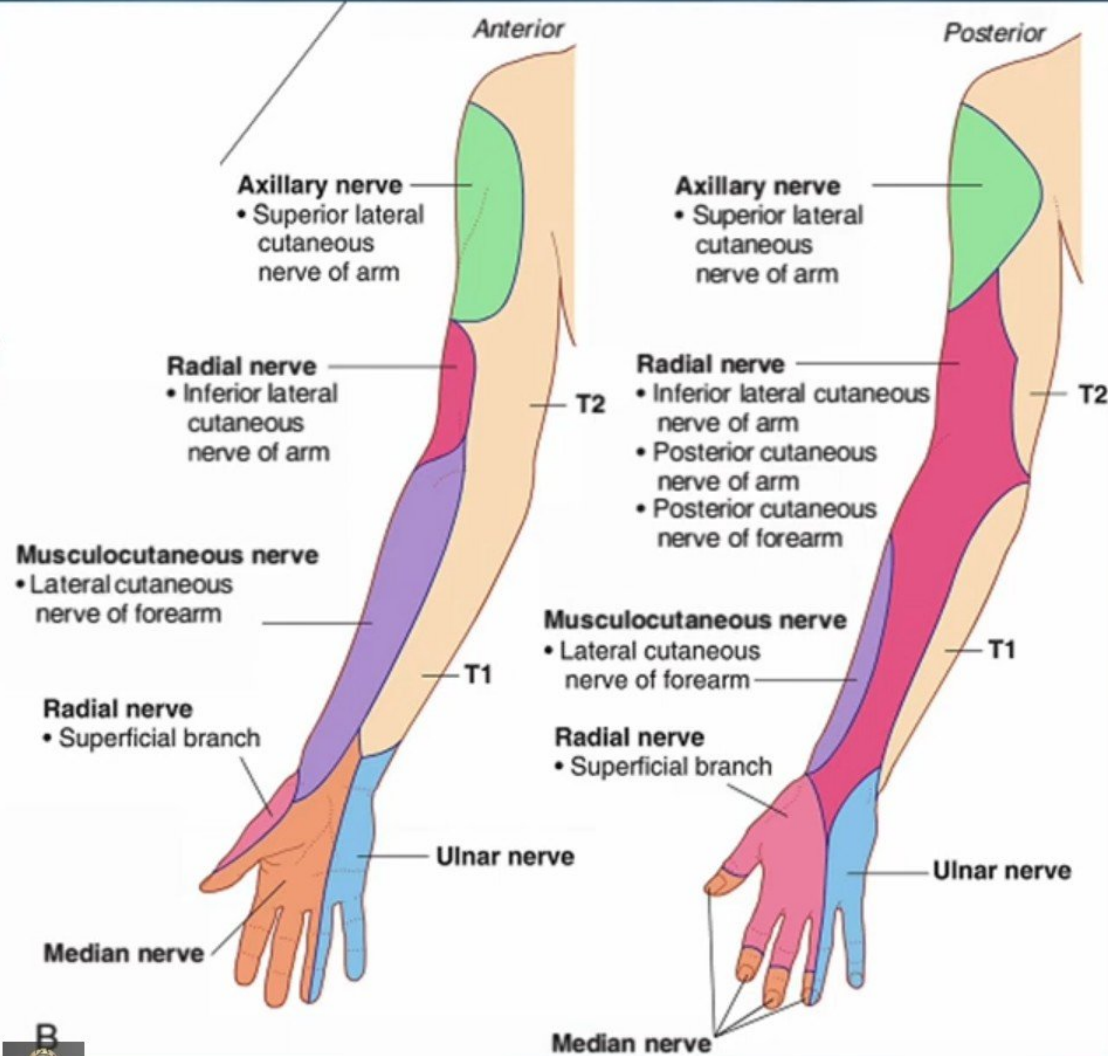
Innervation

- Sensory (dermatome)
- Motor (myotome)



Innervation

- Sensory (dermatome)
- Motor (myotome)



*Thank you &
good luck.. 😊*